CoAdapt-GUI: Joint Workflow Context and Policy Adaptation for Unseen GUI Applications

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Paper version: arXiv 2608.11588v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Success on new tasks within a familiar interface therefore does not imply effective behavior in an unfamiliar application.*

Test-time adaptation on an unseen app can update the **policy**, the **workflow context**, or both.

Motivation: three open gaps

1. **Interface gap.** Unfamiliar elements are hard to ground.
2. **Procedure gap.** Correct actions can still miss workflow and completion checks.
3. **Source drop.** Online RL: +26.1 pts on unseen instances, 8.3 on unseen apps.

Evaluation setup

AW-Generalization · 48 | AndroidWorld Plus · 105 | UI-TARS-7B (S1) | Qwen3-VL-8B (Plus) | Opus 4.7 reflector | 20 rounds per app (Plus) | 5 runs (S1 only)

- **All online** success rate; no offline step accuracy.
- **Held out:** apps unseen in source; new instances (S1) or templates (Plus).

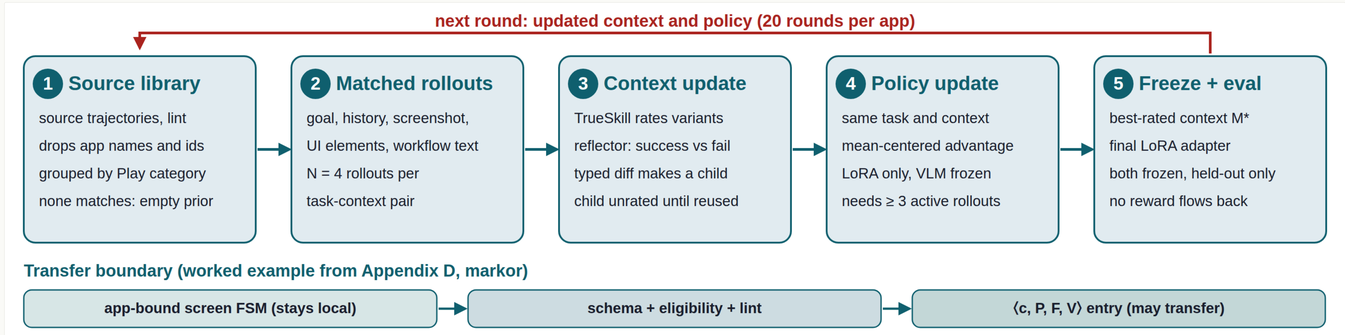
Our read (CUA Radar)

80	★★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; held-out apps; 5 runs (S1); baselines; budget; cost; ablation.
- **Flags:** none recorded. Code link: not stated.
- Pattern C19. Authors' numbers, not rerun.

Method: reward-rated workflow context plus a task–context-matched LoRA

- **Sees / acts:** goal, history, screenshot, UI elements and workflow text in; action out.
- **Context:** typed entry ⟨c, P, F, V⟩; app names, ids and widget labels are linted out.
- **Signal:** executable task reward rates context variants and drives group-relative LoRA updates.

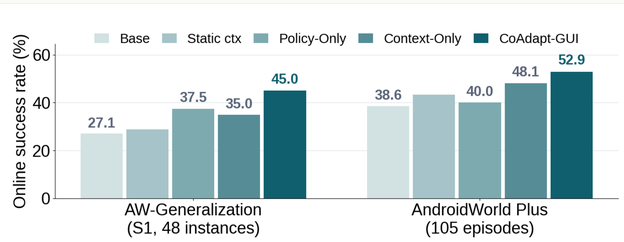


Our redraw (not the paper's figure): one adaptation round, and the transfer boundary from Appendix D.

$$A_j = \frac{r_j - \bar{r}_G}{s_G}, \qquad G(q,\kappa) = \{\, j \mid q_j = q,\ \kappa_j = \kappa \,\}$$

advantage only among rollouts with the same task q and context κ

Key results: online success rate (%)



Tables 1–2, redrawn. S1: Base, Policy-Only reported; others mean of 5 runs (ours). Plus: no std.

45.00 vs 37.50 Policy-Only (S1); 52.9 vs 40.0 (Plus)

SO WHAT → *Base models differ per group.*

Ablation: AndroidWorld Plus overall success (%)

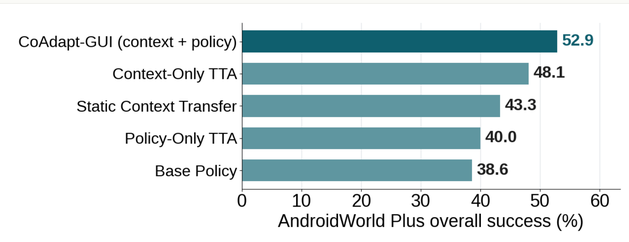


Table 2, redrawn: channel configurations (Qwen3-VL-8B, 105 episodes).

SO WHAT → *Context path 38.6 to 43.3 to 48.1 to 52.9; policy alone 40.0 (+1.4). Category-Novel apps: Policy-Only 25.5, Context-Only 31.4, joint 34.3, Base 29.4. Static retrieval equals Base there (empty source prior).*

Headline Numbers

52.9%
AW Plus overall
vs 38.6% Base

- **45.00%** AW-Gen. (S1)
- **70.4%** Plus, Shared apps
- **34.3%** Plus, Novel apps

The paper's own controls

CONTROL	COADAPT-GUI	COMPARED WITH
S1, 5 runs ± std	45.00 ± 1.86	Context 35.00 ± 1.74
Plus, Shared apps	70.4	Policy 53.7; Ctx 63.9
Plus, Novel apps	34.3	Base 29.4; Policy 25.5
Compute (Plus)	9–10 GPU-h/app	H200 141 GB

Context-Only vs joint: independent runs

Takeaway

Context adaptation helps in both settings; the joint update scores highest, on small held-out panels.

Plus: no std reported | S1 baselines reported | eval covers 12 of 13 apps | no code link stated

Scan to read the full paper

arxiv.org/abs/2608.11588v1